What is a Filter?

Filters are central tools of the UNIX tool kit.

Commands work as follows:

- Accept some data as input.
- Perform some manipulation on the inputted data.
- Produce some output.

Most of them work on a set of records, with each field of a record delimited by a suitable delimiter.

When used in combination, they can perform complex tasks too.

tr Command

The tr command accepts i/p from standard input. This command takes two arguments which specify two character sets.

Converting case ⇒ Lower to Upper and Upper to Lower

who | tr “[a-z]” “[A-Z]”

who | tr “[a-z]” “[A-Z]” > who.upper

```
[student@node1 ~]$
[student@node1 ~]$ cat data1 | tr "[a-z]" "[A-Z]"
I AM DATA1
[student@node1 ~]$
[student@node1 ~]$ cat data1 | tr "[a-z]" "[A-Z]" > upper.txt
[student@node1 ~]$
[student@node1 ~]$ cat upper.txt
I AM DATA1
```

The **–s** option is used to *squeeze* several occurrences of a character to one character.

who | tr -s " "

```
[student@node1 ~]$ who | tr -s " "
root tty1 2024-10-26 22:11
student pts/0 2024-10-26 22:11 (192.168.56.1)
[student@node1 ~]$
[student@node1 ~]$ who | tr -s " " > who.txt
[student@node1 ~]$ cat who.txt
root tty1 2024-10-26 22:11
student pts/0 2024-10-26 22:11 (192.168.56.1)
[student@node1 ~]$
```

cut Command

The cut command retrieves selected fields from a file.

cut [options] <filename>

▪ Options :

• -c : selects columns specified by list

• -f : selects fields specified by list

• -d : field delimiter (default is tab)

tail -n 3 /etc/passwd | cut -d : -f1,2
date | cut -d " " -f1,2

```
[student@node1 ~]$ tail -n 3 /etc/passwd | cut -d : -f1,3,7
chrony:997:/sbin/nologin
systemd-oom:992:/usr/sbin/nologin
student:1000:/bin/bash
[student@node1 ~]$
[student@node1 ~]$ tail -n 3 /etc/passwd | cut -d : -f1,3,7 > users.txt
[student@node1 ~]$ cat users.txt
chrony:997:/sbin/nologin
systemd-oom:992:/usr/sbin/nologin
student:1000:/bin/bash
[student@node1 ~]$
```

sort command

arranges the lines of text from a file.

tail -n 5 /etc/passwd | sort ⇒ Sorts in ascending order

tail -n 5 /etc/passwd | sort -r ⇒ Sorts in descending order

-n ⇒ Numeric Sort

-k ⇒ Sorts via key

-t ⇒ Field separator

tail -n 5 /etc/passwd | sort → ascending order sorting alphabetically

tail -n 5 /etc/passwd | sort -r → reverse sorting alphabetically

tail -n 5 /etc/passwd | sort -t ":" -n -k 3 ⇒ sorts numerically based on the 3rd field

tail -n 5 /etc/passwd | sort -r -t ":" -n -k 3

```
[student@node1 ~]$ tail -n 5 /etc/passwd | sort
chrony:x:997:994::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
sssd:x:998:995:User for sssd:/:/sbin/nologin
student:x:1000:1000::/home/student:/bin/bash
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
[student@node1 ~]$
```

```
[student@node1 ~]$ tail -n 5 /etc/passwd | sort -r
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
student:x:1000:1000::/home/student:/bin/bash
sssd:x:998:995:User for sssd:/:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:997:994::/var/lib/chrony:/sbin/nologin
[student@node1 ~]$
```

```
[student@node1 ~]$ tail -n 5 /etc/passwd | sort -t ":" -n -k 3
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
chrony:x:997:994::/var/lib/chrony:/sbin/nologin
sssd:x:998:995:User for sssd:/:/sbin/nologin
student:x:1000:1000::/home/student:/bin/bash
[student@node1 ~]$
```

```
[student@node1 ~]$ tail -n 5 /etc/passwd | sort -r -t ":" -n -k 3
student:x:1000:1000::/home/student:/bin/bash
sssd:x:998:995:User for sssd:/:/sbin/nologin
chrony:x:997:994::/var/lib/chrony:/sbin/nologin
systemd-oom:x:992:992:systemd Userspace OOM Killer:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
[student@node1 ~]$
```

Uniq command

⇒ this command removes consecutive duplicate entries

⇒ Display the login shells of different users in your system. Shell names should not be duplicated in the output.

⇒ Display the IP Address only for your ethernet interface using the ifconfig command.

⇒ Store the no. Of lines from */etc/passwd* file into passwd.count

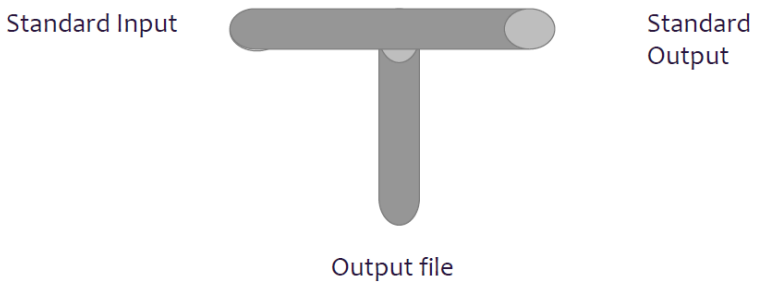
1. cat /etc/passwd | cut -d ":" -f7 | sort | uniq
2. ifconfig enp0s3 | head -2 | grep inet | tr -s " " | cut -d " " -f3

```
[student@node1 ~]$ cat /etc/passwd | cut -d ":" -f7 | uniq
/bin/bash
/sbin/nologin
/bin/sync
/sbin/shutdown
/sbin/halt
/sbin/nologin
/usr/sbin/nologin
/bin/bash
[student@node1 ~]$
```

```
[root@node1 ~]# ifconfig enp0s3 | head -n 2 | grep inet | tr -s " " | cut -d " " -f3
192.168.56.112
[root@node1 ~]# ifconfig enp0s3 | head -n 2 | grep inet | tr -s " " | cut -d " " -f3 > ip.txt
[root@node1 ~]#
```

```
[root@node1 ~]# cat /etc/passwd |wc | tr -s " " | cut -d " " -f2 > passwd.count
[root@node1 ~]#
[root@node1 ~]# cat passwd.count
21
```

tee ⇒ The tee command copies the standard input to the standard output and also to the specified file.



who | tee who.txt ⇒ displays the output on screen and stores it in the specified file also.

Display the list of users who are currently logged in and also count the total no. of logged in users ⇒ who -q

who | tee /dev/pts/1 | wc -l

```
[root@node1 ~]#
[root@node1 ~]# cat /etc/passwd | tee newuser.pdf
```